NOT MEASUREMENT SENSITIVE

MIL-H-40082B
17 September 1992
SUPERSEDING
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30 September 1968

MILITARY SPECIFICATION

HUBS AND DIES FOR HERALDIC ITEMS

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers steel hubs and dies used primarily for extracting working dies used in the manufacture of heraldic items (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

STANDARDS

FEDERAL

FED-STD-151 - Metal, Test Methods

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in preparing this document should be address to: Director, The Institute of Heraldry, US Army, Cameron Station, Alexandria, VA 22304-5050, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document.

AMSC N/A FSC 8455

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(Unless otherwise indicated, copies of federal and military specifications, standards and handbooks are available from: Standardization Document Order Desk, Building 4D, Section D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are issues of the documents cited in the solicitation (see 6.2)

AMERICAN SOCIETY FOR TESTING AND MATERIALS

ASTM-D-3951 - Standard Practice for Commercial Packaging

(Application for copies of ASTM publications should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Non-government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other information services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein (except for related associated detail specifications or MS standards), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Design. The heraldic design of the hub or die shall be an exact replica of the Government furnished drawing, galvano, hub or die as specified (see 6.2).

3.2 Material.

3.2.1 Steel. The steel for hubs and dies shall be a high carbon tool steel, a special purpose steel, or a shock resistant steel suitable for the purpose. Steel designated by the American Iron and Steel Institute (AISI) as carbon tool steels W1, W2, or W4 and one of the following grades: 1, 2, best, special or extra are acceptable. Other special formulations, some of which have not yet been classified by AISI but are commonly used within the industry for this purpose, are acceptable. The following steels listed by their trade name or AISI designation are known to be suitable for the manufacture of hubs and dies.

NAME	C	Mn	Si	Cr	V	Mo	W	REMAINDER
S7	.50	.70	.30	3.25	-	1.40	-	Fe
L7	.70	.55	.30	.85	-	.25	-	Fe
AISI Type S2	.50	.40	1.00	-	-	.50	-	Fe
AISI Type 01 (Ketos)	.90	1.35	.35	.50	-	-	.50	Fe

*These steels may be accepted on the manufacturer's certification.

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Chemical Abbreviations: Carbon (C), Manganese (Mn), Silicon (Si), Chromium (Cr), Vanadium (V), Molybdenum (Mo), Tungsten (W), Iron (Fe)

3.3 Construction. Hubs and dies shall be fabricated from steel specified in 3.2. Unless otherwise specified (see 6.2), the shape of the hubs and dies shall be round, square or oblong. The heraldic design shall be an exact replica of the design furnished by the Government except when required to be of a specified reduced size (see 6.2). Dies for items requiring enamel shall be cut with an overall depth of 0.035 inch (0.089 cm) (normally at dike lines). Areas to be enamelled shall be cut to an overall depth of 0.015 inch + 0.002 inch - 0.000 inch (0.0381 cm + 0.005 cm - 0.00 cm) Pierced areas shall be cut to an overall depth of 0.005 inch + 0.002 inch - 0.000 inch (0.0127 cm + 0.005 cm - 0.00 cm). Modelled areas or detail lines shall be cut lower than the dike lines.

3.3.1 Hubs and die sizes. The minimum height of hubs and dies shall be 2 inches (5.08 cm) for heraldic designs under 4 inches (10.16 cm) in length, width or diameter. The minimum height of hubs and dies for heraldic designs 4 inches (10.16 cm) and over in length, width or diameter shall be 2-1/2 inches (6.35 cm). The minimum length, width or diameter of hubs shall provide a minimum of 3/8 inch (0.95 cm) clearance between the outside edge of the design and the edge of the hub. The minimum length, width or diameter of dies shall provide a minimum of 1/2 inch (1.27 cm) clearance between the outside side of the design and the edge of the die. There shall be sufficient cutaway or relieved area around the embossing on each hub to insure penetration past the embossed design (driving depth) when driving the hub into the working die.

3.3.2 Lead strikes. A lead strike from the die shall be submitted for Government approval and acceptance before the hub or die is hardened. The contractor may substitute an unhardened hub or die for approval in place of a lead strike.

3.3.3 Hardening. Hubs and dies shall be normalized, hardened and tempered. Hardness shall be Rockwell 58 to 63 C scale when tested as specified in 4.4.

3.4 Finish. The top and bottom surfaces of all dies shall be ground parallel. The bottom surface of all hubs shall be ground smooth. The top surface shall have a minimum of three areas spaced 120 degrees apart, ground parallel to the bottom surface for hardness testing.

3.5 Marking for identification. Hubs and dies shall be legibly marked on the outer periphery by engraving or metal stamping with the following information.

a. Type of Steel

b. Date hubs and dies were completed

c. Name or trade mark of manufacturer

d. Words "U.S. Property" or "U.S. Prop" in characters not less than 1/8 inch (0.32 cm) high

3.6 Workmanship. The hubs and dies shall be clean, well made and free from imperfections which may adversely affect their durability and serviceability.

4. QUALITY ASSURANCE PROVISIONS

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4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to the prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspections set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products and supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements; however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Pre-hardening inspection. Inspection of the lead strike or unhardened hub or die specified in 3.3.2 shall be accomplished to determine conformance with the heraldic design requirements specified in 3.3.

4.3 Inspection for acceptance.

4.3.1 Certification of compliance. Components and materials listed below may be accepted on the basis of a contractor's certification of compliance for requirements specified in applicable paragraphs of this specification.

Component	Characteristics	Rqm't Para.
Steel	Material Identification	3.3

4.3.2 Inspection of the end item.

4.3.2.1 Visual examination of hubs and dies. The defect found during examination shall be classified in accordance with 4.3.2.1.1 and 4.3.2.1.2. The hubs and dies shall be inspected in accordance with 4.3.2.1.3. The unit of product for these examinations shall be one hub or die.

4.3.2.1.1 Examination of hubs and dies for defects in design, material, construction and workmanship.

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EXAMINE	DEFECT
DESIGN	Any characteristic not conforming to applicable drawing or other hub or die as specified. Any significant detail not clear, marred, missing, reduced or obliterated; i.e., does not compare favorable with applicable drawing, galvano, hub or die as specified. Heraldic design not an exact replica of the Government furnished design or when required, not of a specified reduced size.
CONSTRUCTION	Not shaped as specified; i.e., not round, square, oblong or as otherwise specified. Driving room on hub omitted.
WORKMANSHIP	Mutilated or not free from imperfections such as scratches, tool marks, dig, gouge, dent, burr, splinter, rough edges, sharp fin or sharp edge which may effect durability or serviceability. Fracture, rupture, crack or broken. Malformed, damaged, warped, twisted, distorted, bent out of shape or otherwise impaired to an extent affecting durability or serviceability.
QUALITY OF METAL	Not fabricated from specified material. Surface spotted or open grained, i.e., scale, corrosion, rust or sludge deposits.
CLEANNESS	Not clean, i.e., scale, corrosion, Bark, rust or sludge deposits.
IDENTIFICATION MARKING	Missing, incorrect, illegible or text incomplete. Words "U.S. Property" or "U.S. Prop" in characters less than 1/8 inch (0.32 cm) high.
FINISH	Surfaces not ground as required.

4.3.2.1.2 Examination of hubs and dies for defects in dimension. Any dimension that is not within specified tolerances shall be classified as a defect.

4.3.2.1.3 Inspection level. One hundred percent inspection shall be made for defects applicable to 4.3.2.1.1, 4.3.2.1.2 and 4.3.2.2. The hub or die shall be rejected if any defect is found.

4.3.2.2 Testing of the end item. Testing of the completed fabricated hub or die shall be performed in accordance with Table I for the characteristics shown therein. The sample unit shall be one completely fabricated hub or die.

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TABLE I End Item Testing

Characteristic	Rqm't Para	Test Method	Rqm't applicable to s/u	#determ per s/u	Results reported as
Hardness 1/	3.3.3	4.4	X	2	To nearest whole number

1/ Care shall be exercised to limit hardness readings to areas where indentations will not be harmful to the heraldic design.

4.4 Tests.

4.4.1 Hardness test. The hardness specified in 3.4.2 shall be determined by the Rockwell Hardness Test, Method 243 in FED-STD-151.

5. PREPARATION FOR DELIVERY

5.1 Preservation. Preservation shall be commercial.

5.1.1 Commercial. Hubs and dies shall be preserved and packaged in accordance with the applicable requirements of ASTM-D-3951.

5.2 Packing. Packing shall be commercial. In addition, the design of each individual hub and die will be capped with wax or protected with corrugated cardboard or similar material before individually wrapped for shipment to the first view station.

5.2.1 Commercial. Hubs and dies preserved as specified in 5.1 shall be packed in accordance with the applicable requirements of ASTM-D-3951.

5.3 Marking. In addition to any special marking required by the contract or order, interior packages and shipping containers shall be marked in accordance with ASTM-D-3951.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended Use. The hubs and dies covered by this specification are intended to be loaned to contractors for extracting their working dies in the manufacture of heraldic items. This specification also covers master dies to be retained by the Government and used for replacement of Government hubs.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.

(b) Whether drawing, galvano, hub or die is to be furnished (see 3.1).

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(c) Shape of hub or die, if not other than specified (see 3.3).

(d) Size of heraldic design when reduction is required (see 3.3).

6.3 Subject term (Key word) listing.

Tool
Steel
Galvano

6.4 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodians:
Army - IH
Air Force - 11
Navy - NU

Preparing Activity:
Army - IH

Review Activities:
Air Force - 82, 99
Navy - MC
DLA - CT

(Project No. 8455-0646)

User Activity:
Air Force - 45

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1 The preparing activity must complete blocks 1, 2, 3, and 8 In block 1, both the document number and revision letter should be given

2 The submitter of this form must complete blocks 4, 5, 6, and 7

3 The preparing activity must provide a reply within 30 days from receipt of the form

NOTE This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements

I RECOMMEND A CHANGE:	1 DOCUMENT NUMBER MIL-L-40088B	2 DOCUMENT DATE (YYMMDD) 920917

3 DOCUMENT TITLE
HUBS AND DIES FOR HERALDIC ITEMS, GENERAL SPECIFICATION FOR

4 NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible Attach extra sheets as needed)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8 PREPARING ACTIVITY

a NAME	b TELEPHONE (Include Area Code) (1) Commercial	(2) AUTOVON
The Institute of Heraldry	(703) 274-6636	284-6636
c ADDRESS (Include Zip Code) Bldg 15, Technical & Production Division Cameron Station Alexandria, VA 22304-5050	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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